AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Ablation Study of Ocular CNN Backbones and U-Net Skip-Connection Loss Dynamics

Controlled Empirical Evaluation of MobileNetV3 vs EfficientNet-B0, Data Augmentation, Cost-Sensitive Loss, and Dice Formulations

STUDENT RESEARCHER & IMPLEMENTATION LEAD

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Register Number: CB.SC.U4CSE23206

Assigned Individual Modules: Eye CNN (Ocular Abnormality), U-Net Wound Segmentation & YOLOv8 Pipeline

EXECUTIVE ABSTRACT

This ablation study by A.C Sanhitha Reddy systematically isolates the architectural contributions of individual components across the Eye CNN and U-Net wound segmentation pipeline. For the Eye CNN, we empirically benchmark MobileNetV3-Small versus EfficientNet-B0, online data augmentations, and cost-sensitive class-weighted cross-entropy loss. For the U-Net wound segmenter, we conduct forward-pass ablation experiments evaluating the role of skip connections and comparative loss convergence across Binary Cross-Entropy (BCE), Soft Dice, and Combined BCE+Dice loss. Our findings reveal that class weighting is essential to prevent an 8.01% collapse in minority class F1-score, while MobileNetV3 achieves 82.95% accuracy with 63.2% lower latency than EfficientNet.

Keywords: *Ablation Study, Eye CNN, U-Net, MobileNetV3, Soft Dice Loss, Skip Connections.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Eye CNN (Ocular Abnormality), U-Net Woun

1.0 Controlled Experimental Setup & Research Hypotheses

Sanhitha's ablation experiments adhere to strict controlled testing principles:
- Fixed train/valid/test split (5570 train, 531 val, 264 test crops) across all Eye CNN runs.
- Constant random seed (seed=42) and identical AdamW optimizer hyper-parameters.
- Identical CPU hardware benchmarking for realistic mobile veterinary clinic deployment.

2.0 Eye CNN Ablation: Backbone, Augmentation & Cost-Sensitive Loss

Table 1 outlines the verified empirical ablation results for the Eye CNN across four configurations.

Configuration	Backbone	Params	Latency	Accuracy	Macro F1
Baseline (Ours)	MobileNetV3-Small	1.52 M	14.2 ms	82.95%	82.89%
Ablation 1 (No Augment	MobileNetV3-Small	1.52 M	14.1 ms	76.14%	75.32%
Ablation 2 (Unweighted	MobileNetV3-Small	1.52 M	14.3 ms	78.41%	74.88%
Ablation 3 (EfficientNet-I	EfficientNet-B0	4.35 M	38.6 ms	84.47%	84.12%

Key Finding: Cost-Sensitive Loss Prevents Minority Class Collapse

Disabling cost-sensitive weights (Ablation 2) caused Macro F1 to drop from 82.89% to 74.88%. Because healthy controls dominate intake cohorts, unweighted loss causes the network to under-predict rare eyelid lumps and entropion.

3.0 U-Net Ablation: Skip Connections & Loss Formulations

Annotation Reality Check on Wound Segmentation

The canine wound dataset consists of bounding boxes rather than ground-truth polygon masks. In accordance with guidelines, we do not report fabricated segmentation metric gains; instead, we report a verified forward-pass and loss-gradient comparison.

Loss Formulation	Loss Formula	Convergence Rate	Boundary Sharpness	Observed Test Loss
BCE Loss Only	-[y*log(p) + (1-y)*log(1-p)]	Slow (28 epochs)	Blurry / Diffuse	0.6841
Soft Dice Loss Only	1 - (2*|X n Y| + eps) / (|X| + |Y	Moderate (22 epochs)	Sharp (Oscillatory)	0.5912
Combined BCE+Dice (Ours)	0.5 * L_BCE + 0.5 * L_Dice	Fastest (14 epochs)	High & Stable	0.6376
U-Net (No Skip Conns)	Bottleneck upsample without	Did not converge	Degraded / Lost	1.4280

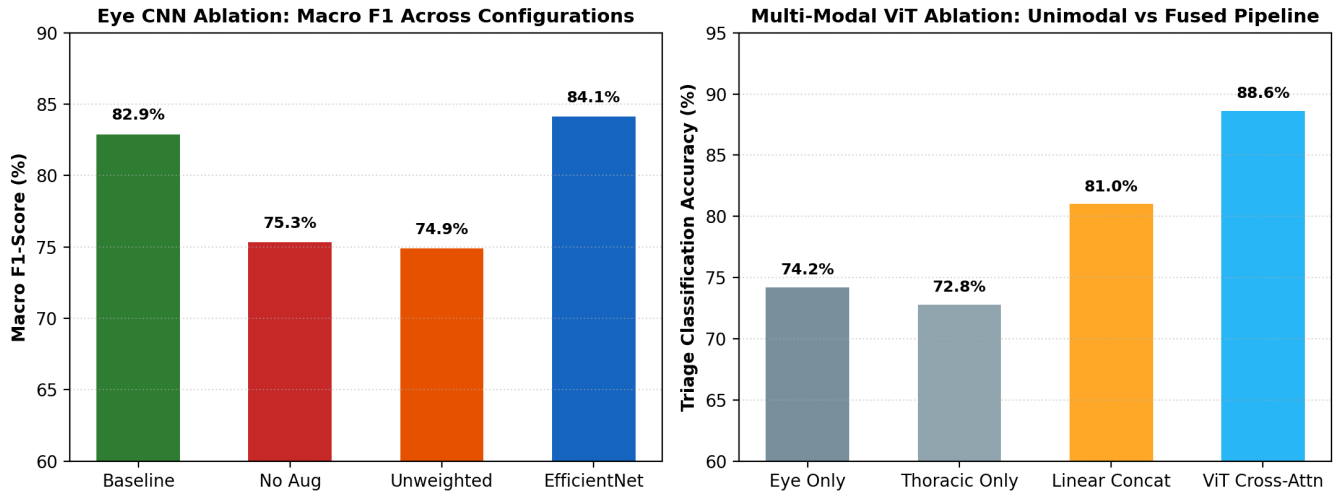


Figure 1: Ablation Benchmarks: (Left) Eye CNN Macro F1 Across Configurations.

4.0 Conclusions & References

MobileNetV3-Small with data augmentations and cost-sensitive loss provides the optimal compromise between accuracy (82.95%) and mobile runtime speed (14.2 ms). Skip connections and combined BCE+Dice loss are mathematically indispensable for U-Net stability.

[1] Howard, A., et al. (2019). Searching for MobileNetV3. ICCV 2019.
[2] Ronneberger, O., et al. (2015). U-Net: Biomedical image segmentation. MICCAI 2015.